

Deep reinforcement learning-based device association in hybrid LiFi/WiFi indoor IoT networks

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ABSTRACT

Hybrid light fidelity (LiFi) and wireless fidelity (WiFi) indoor networks have been envisioned as a promising technology to alleviate the radio frequency spectrum crunch to accommodate the ever-increasing data rate demand in indoor internet of things (IoT) networks. The hybrid LiFi/WiFi indoor IoT networks can leverage the advantages of fast data transmission from LiFi and wider coverage of WiFi, thus complementing well with each other and further improving the network performance compared with the standalone networks. However, to leverage the co-existence, several challenges should be addressed, including but not limited to device association, mobility support, and LiFi AP blockage response. Therefore, the objective of the paper is to design a new device-access point association algorithm to maximize the sum throughput of the hybrid networks. We first mathematically formulate the sum data rate maximization problem by determining the AP selection for each IoT device in indoor networks with consideration of IoT device mobility and practical capacity limitations, which is a nonconvex binary integer programming problem. To solve this problem, we then propose a sequential-proximal policy optimization (S-PPO) based deep reinforcement learning method. Extensive simulations are conducted to evaluate the proposed method by comparing it with exhaustive search (ES), linear programming rounding (LPR), signal strength strategy (SSS), trust region policy optimization (TRPO), and TRPO-domain knowledge (DK) methods. Comprehensive simulation results demonstrate that our solution algorithm can outperform SSS by about 32.3 % of the sum throughput and 19.1 % of the fairness on average, outperform TRPO by about 10.3 % and 10.2 %, and outperform TRPO-DK by about 16.8 % and 18.3 %, respectively.

1. Introduction

Over the past few years, the number of internet of things (IoT) devices has increased dramatically, driven by the increasing demand for smart homes, industrial automation, healthcare monitoring, and so on. According to the latest data, the number of IoT devices worldwide is expected to exceed 39.2 billion by 2029 [1]. The increasing number of these IoT devices, especially in confined areas of indoor environments, will inevitably further congest the traditional radio frequency (RF) spectrum. To address the challenges, visible light communication (VLC), where light-emitting diodes (LEDs) and photodetectors (PDs) are used to transmit and receive data using the optical spectrum, has been envisioned as a promising solution because of its license-free orders of magnitude of available bandwidth, enhanced security, electromagnetic interference-free nature, and easy integration with existing infrastructure [2–7]. In recent years, light fidelity (LiFi) has been proposed, which utilizes VLC for data transmission, and can be installed on existing lighting infrastructures, thus realizing a dual-purpose system that

provides illumination and communication simultaneously. Unlike wireless fidelity (WiFi) technology, which operates on the crowded radio frequency (RF) spectrum, typically in the 2.4 GHz and 5 GHz bands, LiFi utilizes a large portion of the unregulated spectrum ranging from 400 to 800 THz, thereby offering nearly 10,000 times more available bandwidth. The main advantages of LiFi technology include: (i) high-speed data transmission potential relying on the spectrum of orders of magnitude [8]; (ii) safe in RF-restricted environments because of no electromagnetic interference to RF, such as aircraft and hospitals; (iii) inherently secure due to the low-penetration property [3]. Though LiFi offers significant advantages, it also has limitations: (i) limited field of view (FoV), usually a few meters for a single AP; (ii) sensitive to obstacles due to low penetration [9].

Integrating LiFi with existing WiFi networks to form hybrid LiFi/WiFi networks provides a solution that leverages the advantages of fast data transmission of LiFi and wider coverage range of WiFi, thus complementing well with each other to provide a seamless, high data-rate transmission in indoor networks [10]. Recently, a number of studies

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have been conducted with the objective of improving the spectrum efficiency of hybrid LiFi/WiFi networks. The authors in [11] propose a Q-learning-based dual-timescale power allocation approach for multi-homing hybrid RF/VLC networks, aiming at meeting the quality of service (QoS) demands of users. A learning-based solution for optimizing network selection, subchannel distribution, and power allocation is proposed in [12] to satisfy the diverse demands of users in heterogeneous RF/VLC networks. The work in [13] explores the topology control problem in a hybrid VLC/RF vehicular ad-hoc network, and proposes a distributed algorithm. A fuzzy logic-based dynamic handover scheme for indoor LiFi and RF hybrid networks is proposed in [14] to reduce the handover overhead. Among these efforts, network selection plays a crucial role in enhancing spectrum efficiency. However, traditional device-access point (AP) selection methods for homogeneous networks, such as signal strength strategy (SSS) [15] are not applicable as they will make the WiFi APs prone to overloading.¹ Moreover, the dynamic characteristics of IoT device requirements and mobility in indoor heterogeneous networks make the device-access point association problem more challenging.

In recent years, a few studies [16–25] have been conducted to propose device-AP association methods specifically for the hybrid LiFi/WiFi indoor networks. However, most traditional methods, such as game theory [22,23], and fuzzy logic-based AP selection [24], do not consider the mobility of IoT devices, which is critical for real-world applications such as smartwatches, where mobile devices frequently move and experience dynamic connectivity. Moreover, conventional methods often assume that channel state information (CSI) is perfectly known. However, in practical applications, CSI is typically unpredictable and susceptible to interference. To effectively address the dynamic and complex decision-making challenges posed by device mobility in wireless environments, adopting learning-based methods emerges as an effective solution [19,20,25]. Recently proposed supervised learning methods [25] require a large set of training datasets, which is difficult to obtain in practical wireless networks. In contrast, deep reinforcement learning (DRL) can learn optimal behavior through continuous trial and error by interacting with the time-varying networking environments without a strict requirement of collecting a large volume of training datasets, thus providing a more efficient approach to optimizing device association that adapts as the network changes. Yet, current DRL works in [19,20] do not consider the practical capacity limitations of APs, which makes the scenario unrealistic.

To bridge the gap, our objective is to propose a robust device-AP association method for indoor hybrid LiFi/WiFi networks with either static or mobile devices, aiming to maximize the achievable sum data rate. Due to the binary nature of the device-AP association variables, the resulting formulated problem is classified as a nonconvex binary integer programming (BIP) problem, typically NP-hard, implying that finding a globally optimal solution within polynomial time is unfeasible. Therefore, we propose a sequential-proximal policy optimization (S-PPO) based solution method to determine the device-AP association. The main contributions of this paper are outlined as follows:

- *Device-AP association formulation.* We mathematically formulate the device-AP association problem for indoor hybrid LiFi/WiFi networks with the objective of maximizing the sum throughput by jointly considering the network capacity limitation and the device mobility patterns.
- *Sequential-proximal policy optimization based solution algorithm.* To solve the formulated binary integer programming problem, we propose a solution algorithm based on proximal policy optimization, an advanced reinforcement learning framework that combines the benefits of both actor-critic and policy gradient methods. Moreover, we

design an action space decomposition strategy that significantly reduces the action space by decomposing the entire association policy into a sequencing policy.

- *Performance evaluation.* We evaluate the performance of the proposed S-PPO method with respect to optimality, convergence, robustness, computational complexity, scalability, and fairness in three network scenarios and three blockage scenarios. We also compare the proposed S-PPO with exhaustive search (ES), linear programming rounding (LPR), signal strength strategy, trust region policy optimization (TRPO), and TRPO-domain knowledge (DK) methods through extensive simulation experiments. Comprehensive simulation results show that the proposed S-PPO outperforms the baseline methods in terms of sum data rate, fairness, and runtime under different network topologies and capacity limitations.

The remainder of the paper is organized as follows. Section 2 reviews the state-of-the-art. In Section 3, we introduce the system model for the hybrid LiFi/WiFi networks, the device-AP association model, and the random waypoint model. The centralized optimization problem formulation is present in Section 4. Subsequently, we discuss the proposed S-PPO method in Section 5. Lastly, Section 6 evaluates the performance of the proposed method, followed by an analysis of limitations and future direction in Section 7, with conclusions highlighted in Section 8.

2. Related work

2.1. Optimization based methods

There are some efforts that have been made in the literature to solve the device-AP association problem in hybrid LiFi/WiFi networks [16,19,20,24,26]. In Wu et al. [24], a two-stage fuzzy logic-based system is proposed to assign users to APs. The proposed system calculates the accessibility scores of users and then assigns users to WiFi APs in descending order of scores until the maximum capacity of WiFi APs is reached. The remaining users are then served by LiFi APs. With this approach, the throughput of the system has been greatly improved. The authors in [16] design a data rate threshold in hybrid LiFi/WiFi networks, where all the devices first connect to LiFi APs, and those below the threshold switch to WiFi APs. The simulation results show that the hybrid networks have higher spectral efficiency than the stand-alone network. The authors in [26] propose a solution that utilizes the individual throughput of each station and overall network energy consumption to determine the association between stations and APs. Compared with baseline methods, the proposed approach achieves better trade-offs in terms of throughput and energy efficiency.

However, these studies assume that devices remain stationary, which does not align with the dynamics of devices in the real world. Furthermore, the interference among LiFi APs is not considered in [26], which makes it lack generality. Different from previous works, we take into account device mobility and inter-cell interference for LiFi APs. As devices move, channel information changes, thereby complicating the device-AP association problem.

2.2. Reinforcement learning based methods

Reinforcement learning (RL) based methods have attracted considerable attention from researchers for their potential to solve the device-AP association problem in hybrid LiFi/WiFi networks. In [19], a Q-learning with knowledge transfer method is proposed to enable online networking selection, which outperforms the traditional Q-learning model and has a faster convergence speed. However, the method relies on knowledge or contextual information, which may vary across different network scenarios, and sometimes the contextual information is not available. The work in [20] implements trust region policy optimization to provide an optimal device association strategy with consideration of different device mobility patterns. The results show that the proposed RL

¹ In hybrid networks, WiFi has a larger coverage area but lower capacity compared with LiFi, thus WiFi AP will be likely to overload when SSS based approaches are adopted.

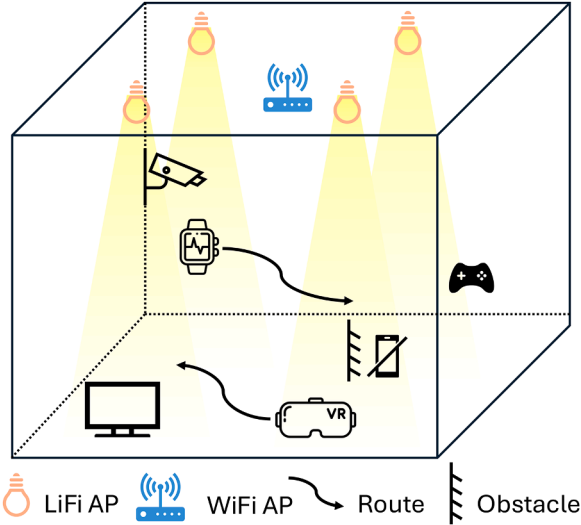


Fig. 1. Indoor hybrid LiFi/WiFi networks.

method outperforms the iterative algorithm and signal strength strategy method in most cases. In [27], a TRPO-based method is also proposed for load balancing in heterogeneous LiFi-WiFi networks. However, the TRPO-based method used in this work is inefficient in terms of computational expenses and sample efficiency due to second-order optimization.

Inspired by the above observations, we proposed a fast-converging and computationally efficient sequential proximal policy optimization-based actor-critic reinforcement learning algorithm to tackle the device-AP association problem in hybrid LiFi/WiFi indoor networks, taking into account the mobility of devices as well as network capacity constraints.

3. System model

We consider an indoor hybrid LiFi/WiFi IoT network as illustrated in Fig. 1. The indoor scenario consists of a set of $\mathcal{L} = \{1, 2, \dots, l, \dots, L\}$ LiFi APs and a set of $\mathcal{W} = \{1, 2, \dots, w, \dots, W\}$ WiFi APs. We assume that there is a set of $\mathcal{K} = \{1, 2, \dots, k, \dots, K\}$ active devices randomly distributed over the room, where devices can only be served by either one LiFi AP or one WiFi AP. We assume each AP adopts time division multiple access (TDMA) to control the channel access for multiple users with predefined time slots. From a practical perspective, driven by the finite capacity limitation of the networks, we also limit the maximum number of connected devices to each LiFi AP and WiFi AP as N_l and N_w as in [28].

3.1. LiFi channel model

In this study, we consider the line of sight (LoS) downlink path in LiFi networks. Our following analysis can be easily extended to scenarios with LoS and non-LoS paths. The geometry of the LiFi LoS propagation is shown in Fig. 2, where the LED transmitter – LiFi AP l is installed on the ceiling, directly facing downwards, and the receiving device photodiode (PD) is installed on each device k , facing directly upwards. $d_{l,k}(t)$, $\psi_{l,k}(t)$, and $\phi_{l,k}(t)$ are the distance between the LiFi AP l and the device k , the angle of incidence, and the angle of irradiation at time slot t , respectively. $\Psi_{1/2}$ is the half angle of the PD's FoV, and $\Phi_{1/2}$ is the half-intensity radiation angle of the LiFi AP. At time slot t , if device k is served by LiFi AP l , the optical channel gain of the LoS channel can

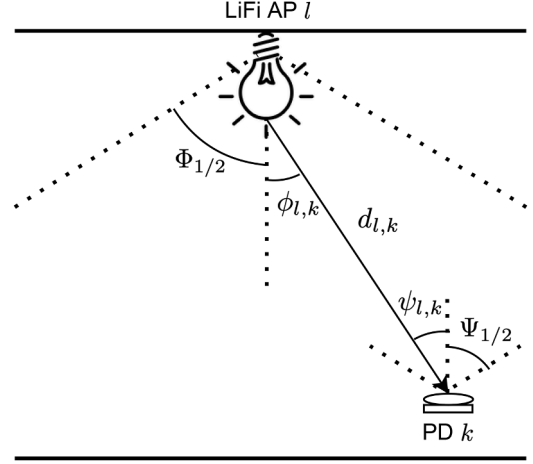


Fig. 2. LiFi LoS channel model.

then be given as [29]:

$$H_{l,k}^{lf}(t) = \begin{cases} \frac{(m+1)A}{2\pi d_{l,k}^2(t)} \cos^m(\phi_{l,k}(t)) T_s(\psi_{l,k}) g(\psi_{l,k}(t)) \cos(\psi_{l,k}(t)) & 0 \leq \psi_{l,k}(t) \leq \Psi_{1/2}, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where m is the Lambertian index and is given by the semi-angle $\Psi_{1/2}$ at half illuminance power of the LED as $m = -\ln 2 / \ln(\cos \Psi_{1/2})$. A is the physical area of the PD. $T_s(\psi_{l,k})$ denotes the gain of an optical filter. $g(\psi_{l,k}(t))$ is the optical concentrator gain at time slot t , given as:

$$g(\psi_{l,k}(t)) = \begin{cases} \frac{n^2}{\sin^2 \Psi_{1/2}} & 0 \leq \psi_{l,k}(t) \leq \Psi_{1/2}, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where n is the refractive index. When device k is connected to LiFi AP l at time slot t , the signal-to-interference-plus-noise ratio (SINR) is calculated as:

$$\gamma_{l,k}^{lf}(t) = \frac{(\kappa H_{l,k}^{lf}(t) P_{lf})^2}{\mathcal{N}_{lf} B_{lf} + \sum_{j \neq l} (\kappa H_{j,k}^{lf}(t) P_{lf})^2}, \quad (3)$$

where κ denotes the optical to electric conversion efficiency of the adopted PD; $H_{l,k}^{lf}(t)$ is the channel gain between the device k and the connected LiFi AP l at time slot t ; P_{lf} and B_{lf} are the transmitted optical power and modulation bandwidth of LiFi APs, respectively; $\mathcal{N}_{lf}$ represents the power spectral density of LiFi noise; $H_{j,k}^{lf}(t)$ is the channel gain between the device k and the interfering LiFi AP j at time slot t . Since the signals in LiFi are non-negative, Shannon capacity cannot be used directly. Instead, we use the lower bound in [30], the achievable data rate between LiFi AP l and device k at time slot t can then be given as:

$$r_{l,k}^{lf}(t) = \frac{B_{lf}}{2} \log_2 \left(1 + \frac{e}{2\pi} \gamma_{l,k}^{lf}(t) \right). \quad (4)$$

3.2. WiFi channel model

For WiFi networks, the path loss model defined in [31] is adopted, where free space loss and shadow fading loss due to large-scale obstructions are considered, given as:

$$L(d_{w,k}(t)) = \begin{cases} L_{FS}(d_{w,k}(t)) + X_{SF} & d_{w,k}(t) \leq d_{BP}, \\ L_{FS}(d_{w,k}(t)) + X_{SF} + 35 \log_{10} \left(\frac{d_{w,k}(t)}{d_{BP}} \right) & d_{w,k}(t) > d_{BP}, \end{cases} \quad (5)$$

where $d_{w,k}(t)$ is the distance between the WiFi AP w and the device k at time slot t ; d_{BP} is the breakpoint distance; X_{SF} denotes the shadow fading loss, which is a Gaussian random variable with zero mean and 3 dB deviation if the distance is smaller than or equal to the breakpoint, and 5 dB deviation otherwise; the free space loss L_{FS} is calculated as:

$$L_{FS}(d_{w,k}(t)) = 20\log_{10}(d_{w,k}(t)) + 20\log_{10}(f_c) - 147.5, \quad (6)$$

with f_c denoting the central carrier frequency. At time slot t , the channel transfer function between device k and WiFi AP w can be expressed as:

$$H_{w,k}^{wf}(t) = \sqrt{10^{\frac{-L(d_{w,k}(t))}{10}}} \left(\sqrt{\frac{K}{1+K}} e^{j\phi} + \sqrt{\frac{1}{1+K}} X \right), \quad (7)$$

with K representing the Rician K -factor [31]. For the LoS path, $K = 1$ if $d_{w,k}(t) \leq d_{BP}$, otherwise $K = 0$. ϕ is the angle of arrival or departure of the LoS component. X is a complex Gaussian random variable with zero mean and unit variance. Then, the SINR $\gamma_{w,k}^{wf}(t)$ for the WiFi channel model at time slot t is given as:

$$\gamma_{w,k}^{wf}(t) = \frac{|H_{w,k}^{wf}(t)|^2 P_{wf}}{\mathcal{N}_{wf} B_{wf}}, \quad (8)$$

where $\mathcal{N}_{wf}$ represents the power spectral density of noise of the WiFi networks; P_{wf} and B_{wf} are the transmitted power and modulation bandwidth of each WiFi AP, respectively. The achievable data rate $r_{w,k}^{wf}(t)$ between the WiFi AP w and device k can be calculated using:

$$r_{w,k}^{wf}(t) = \frac{B_{wf}}{2} \log_2(1 + \gamma_{w,k}^{wf}(t)). \quad (9)$$

3.3. Device-AP association model

We consider single-home accommodation for devices in favor of tractable complexity in modeling and theoretical analysis, i.e., each device is associated with at most one AP, either a WiFi AP or a LiFi AP. The device association vector $\mathbf{u}_k(t)$ for device k at time slot t is denoted as $\mathbf{u}_k(t) = \{u_{l,k}(t), u_{w,k}(t) | \forall l \in \mathcal{L}, \forall w \in \mathcal{W}\}$, where $u_{l,k}(t)$ and $u_{w,k}(t)$ are binary variables, with $u_{l,k}(t) = 1$ or $u_{w,k}(t) = 1$ when device k is associated with LiFi AP l or WiFi AP w , respectively, 0 otherwise. Then, we can define the device-AP association vector for the entire hybrid network as $\mathbf{u}(t) = \{\mathbf{u}_k(t) | \forall k \in \mathcal{K}\}$.

Once the connections between the devices and APs are determined, the sum data rate of the hybrid networks at time slot t can be written as:

$$r_t = \sum_{l=1}^{\mathcal{L}} \sum_{k=1}^{\mathcal{K}} u_{l,k}(t) r_{l,k}^{lf}(t) + \sum_{w=1}^{\mathcal{W}} \sum_{k=1}^{\mathcal{K}} u_{w,k}(t) r_{w,k}^{wf}(t). \quad (10)$$

3.4. Random waypoint model

In this work, we use the random waypoint (RWP) model [32] to model devices' indoor moving trajectories, as it effectively replicates users' motion patterns across a range of indoor environments. This model is extensively utilized in past research [17,20]. Within the framework of the RWP model, devices select a destination point at random within the designated area and proceed towards it in a direct line. The speed of each device is also randomly determined, adhering to a random distribution within a specified range of minimum and maximum velocities. Upon reaching the destination, the device may either pause for a set duration or resume movement, adhering to the previously described mechanism. This model accommodates the coexistence of mobile and stationary devices within the environment at any given moment.

3.5. Blockage model

Because of the susceptibility of LiFi AP to blockage caused by opaque obstacles, the LoS path between LED transmitters and PD receivers may

not always be available. Therefore, we design a blockage model to evaluate the impact of both static and dynamic blockage in the proposed indoor scenario [33,34].

• **Static blockage.** The static blockage is proposed to simulate the permanent obstacles in an indoor scenario, such as walls or furniture, where the LoS path between AP l and user k is unavailable when the user is located in the shadowed area caused by these obstacles.

• **Dynamic blockage.** The time-varying blockage is designed to mimic dynamic obstacles in indoor scenarios, such as mobile humans. Each user k is assigned a randomly generated blockage probability $\Pr(i, k)$ between itself and LiFi AP l . Let $\rho_{l,k}$ represent the corresponding binary blockage indicator. The binary indicator will be set to 1 when the blockage probability $\Pr(l, k)$ is greater than a predefined blockage rate $\Pr_{th}$, indicating an unobstructed LoS path, otherwise 0, indicating a blocked path. The blockage model can be represented as:

$$\rho_{l,k} = \begin{cases} 1 & \Pr(l, k) \geq \Pr_{th} \text{ (unblocked)}, \\ 0 & \Pr(l, k) < \Pr_{th} \text{ (blocked)}, \end{cases} \quad (11)$$

It is worth noting that the blockage only occurs when the LiFi AP and user are within each other's FoV, and both $\Pr(l, k)$ and $\rho_{l,k}$ are time-varying parameters.

• **Hybrid blockage.** The hybrid blockage scenario combines both static and dynamic blockage to explore the impact of their coexistence.

4. Problem formulation

The network control objective can be stated as maximizing the sum data rate of the hybrid indoor LiFi/WiFi downlink access networks by jointly considering the devices' positions, device-AP association variables, as well as device moving patterns, subject to bandwidth limitations of the hybrid networks. Define $\mathcal{P}(t) = [\mathcal{P}_1(t), \mathcal{P}_2(t), \dots, \mathcal{P}_K(t)]$ as the device location at time slot t , where height is assumed constant for all devices and only x coordinates and y coordinates are considered to be moving over time, with $\mathcal{P}_k(t) = \{x_k(t), y_k(t)\}$. The network control problem can then be formulated as:

Problem 1:

$$\text{Maximize}_{\mathbf{u}(t)} \quad f = r_t \quad (12)$$

$$\text{Subject to: } u_{l,k}(t) \in \{0, 1\}, \forall k \in \mathcal{K}, \forall l \in \mathcal{L}, \quad (13)$$

$$u_{w,k}(t) \in \{0, 1\}, \forall k \in \mathcal{K}, \forall w \in \mathcal{W}, \quad (14)$$

$$\mathbf{1}^T \mathbf{u}_k(t) \leq 1, \forall k \in \mathcal{K}, \quad (15)$$

$$\sum_{k \in \mathcal{K}} u_{l,k}(t) \leq N_l, \forall l \in \mathcal{L}, \quad (16)$$

$$\sum_{k \in \mathcal{K}} u_{w,k}(t) \leq N_w, \forall w \in \mathcal{W}. \quad (17)$$

In the above-formulated problem, constraints (13) and (14) require that the device-AP association variables are binary. (15) indicates that each device can be served by only one AP. The capacity limitations (aka the maximum number of connected devices) of LiFi and WiFi APs are set in (16) and (17). We consider maximizing the sum throughput in the objective function; however, without loss of generality, our method can be extended to incorporate additional factors such as energy consumption, latency, or fairness.

Corollary 1. Problem 1 is a non-convex binary integer programming (BIP) problem.

Proof. From the constraints (13) and (14), the decision variables $u_{l,k}(t)$ and $u_{w,k}(t)$ are binary. Moreover, r_t in the objective function (12) is the summation of individual throughput terms $r_{l,k}^{lf}$ and $r_{w,k}^{wf}$ in (4) and (9), which are independent of variables $u_{l,k}(t)$ and $u_{w,k}(t)$. Thus, the objective function r_t is a weighted linear summation of binary variables. Additionally, the single-home assignment constraint (15) and the capacity limitation constraints (16) and (17) are linear. Therefore, Problem 1 is classified as a binary integer programming problem [35]. Moreover, the

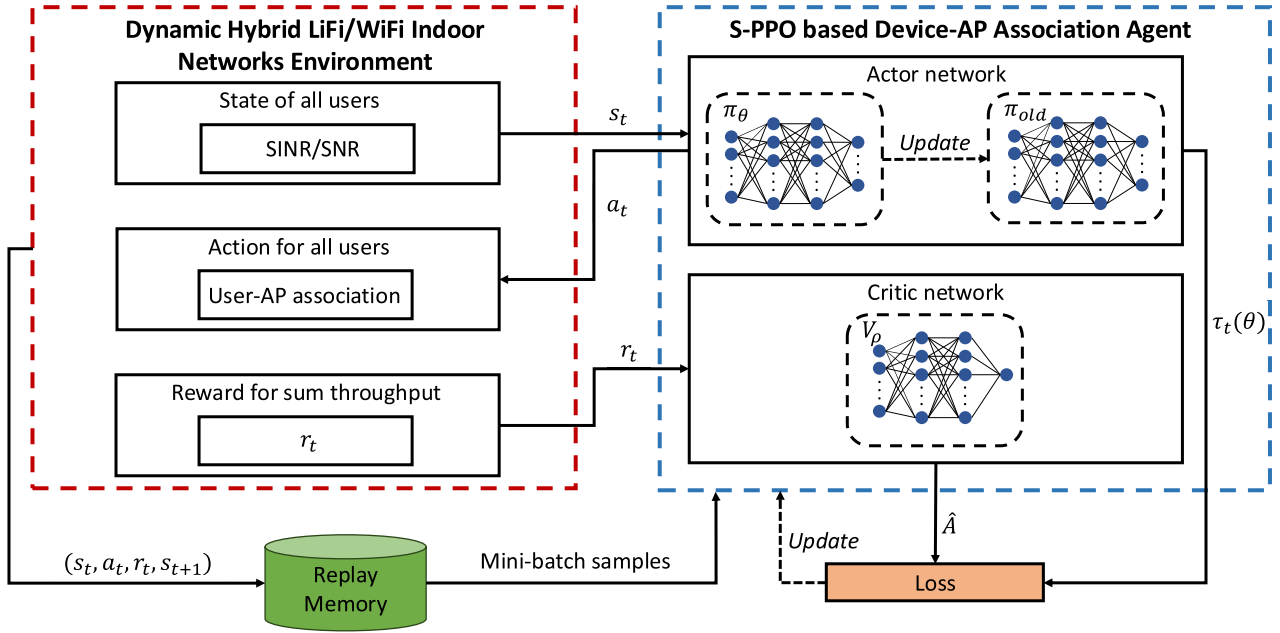


Fig. 3. Actor-Critic S-PPO based device association scheme.

feasible set of Problem 1 is a non-convex set due to binary constraints, thereby we conclude that this BIP problem is non-convex. $\square$

5. Proposed S-PPO method

Problem 1 is a non-convex BIP problem, for which there are generally no existing solution algorithms that can obtain the global optimum in polynomial computational time. Traditional solution algorithms face significant challenges in solving this problem because they are computationally intensive due to the dynamic features of the wireless environment and therefore cannot effectively obtain the global optimal solution. In contrast, learning-based methods can reformulate the optimization challenge into a decision-making problem, showing greater flexibility in dynamic environments. Therefore, in this paper, we propose a sequential-proximal policy optimization method as a solution algorithm for the formulated NP-hard problem. Specifically, the objective function (12) of the BIP is used as the reward to guide policy learning, while AP capacity constraints (16) and (17) are incorporated through penalty terms when violated. Furthermore, an action space decomposition method is designed to significantly reduce computational complexity and accelerate convergence.

5.1. Preliminaries

In this section, we introduce the preliminaries of the proximal policy optimization algorithm. The PPO algorithm is based on the actor-critic framework, which integrates the strengths of both value-based (such as Q-learning) and policy-based (such as gradient descent) approaches in reinforcement learning. This integration enables quicker convergence and more stable performance. Additionally, PPO is less sensitive to the changes in hyperparameters, which makes the tuning process more efficient, thus allowing our proposed PPO-based solution method can be seamlessly applied to various network topologies.

In the proposed S-PPO, the actor-critic framework has two networks, as shown in Fig. 3, the policy network π_{θ_t} (actor) and the value network V_{ρ_t} (critic), where θ_t and ρ_t are the parameters of these two networks at time slot t [36]. Given a state s_t , the probability of the policy π_{θ_t} executes action a_t denoted as $\pi_{\theta_t}(a_t|s_t)$. The underlying principle of the PPO algorithm is to update the policy in a manner that maximizes expected

reward without drastic policy changes, which is achieved by the clipped surrogate objective function [37]:

$$L^{CLIP}(\theta) = \mathbb{E}_t[\min(\tau_t(\theta)\hat{A}_t, \text{clip}(\tau_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)], \quad (18)$$

where $\tau_t(\theta)$ is the probability ratio between the new and old policies. This ratio measures how the new policy deviates from the old one, which can be calculated as:

$$\tau_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}. \quad (19)$$

Moreover, $\hat{A}_t$, the generalized advantage estimator, is used to measure how much better it is to take a specific action compared to the average action at a given state, as:

$$\hat{A}_t = R_t - V_{\rho}(s_t), \quad (20)$$

with R_t representing the accumulative reward, $V_{\rho}(s_t)$ denoting the expected return of the state s_t .

Additionally, the clip function is defined as:

$$\text{clip}(\tau_t(\theta), 1 - \epsilon, 1 + \epsilon) = \begin{cases} 1 - \epsilon & \tau_t(\theta) < 1 - \epsilon, \\ 1 + \epsilon & \tau_t(\theta) > 1 + \epsilon, \\ \tau_t(\theta) & \text{otherwise,} \end{cases} \quad (21)$$

where ϵ is a clipping hyperparameter.

On the other hand, the loss function to update the critic network is typically a mean squared error (MSE) loss, which can be given as:

$$L^{Critic}(\rho) = \frac{1}{N} \sum_{i=1}^N (V_{\rho}(s_i) - R_i)^2, \quad (22)$$

where N is the number of state-value pairs in the batch used for the update.

In summary, at state s_t , the actor network is used to select actions a_t based on the current policy π_{θ} to interact with the environment, and then receive reward r_t . Next, the experience tuples $\{s_t, a_t, r_t, s_{t+1}\}$ will be stored in the memory. After several timesteps, the data in the memory will be used to calculate the loss. The actor and critic network will then be updated by the gradient of $L^{CLIP}(\theta)$ and $L^{Critic}(\rho)$, respectively, thus obtaining the optimal policy.

5.2. Sequential-proximal policy optimization

Next, we present the framework of the proposed RL method and action space decomposition method in detail as follows.

Agent: The agent is a central unit that can obtain the device positions and execute the proposed network control algorithm to accomplish the device-AP association.

Sequential actions: In RL, the high dimensionality of the action space will have a negative impact on sample efficiency and model expressiveness [38]. The potential action space, denoted as $K^{(L+W)}$, expands exponentially with the growth of the number of devices K , and the numbers of APs L and W , respectively. To address the issue of this exponentially increasing action space, we design a new action space decomposition approach. The key idea is to divide the action space into K sequential actions for each device. This approach effectively breaks down the overall association policy into a sequencing policy to determine the action for each device sequentially. After all device-AP association actions are collected, they are composed into a combined action and then sent to the wireless network environment. In this way, the overall action space is reduced to $K(L+W)$.

The action taken at time slot t for each device is determined by the current policy network, expressed as:

$$a_t^k = [\mathbf{u}_k(t)], a_t^k \in \mathcal{A}, k \in \mathcal{K}, \quad (23)$$

which is the device-AP association vector of the hybrid LiFi/WiFi networks for device k . $\mathcal{A}$ is the action space including all possible actions. Then the combined action for all the devices is defined as:

$$\mathbf{a}_t = [\mathbf{u}(t)]. \quad (24)$$

States: To obtain the individual device action, the state is defined as the received SINR/SNR from LiFi/WiFi APs of each device:

$$s_t^k = [\gamma^k(t)], s_t^k \in \mathcal{S}, k \in \mathcal{K}, \quad (25)$$

where $\gamma^k(t) = \{\gamma_{l,k}^{lf}(t), \gamma_{w,k}^{wf}(t)\}$, $l \in \mathcal{L}$, $w \in \mathcal{W}$, and $\mathcal{S}$ denotes the state space. The combined state for all devices can be given by:

$$\mathbf{s}_t = [\gamma(t)]. \quad (26)$$

Policy: The stochastic policy $\pi_\theta(a_t^k | s_t^k)$ is the probability of choosing action a_t^k at state s_t^k , which can be defined as:

$$\pi_\theta(a_t^k | s_t^k) = \Pr(a_t^k | s_t^k). \quad (27)$$

By selecting the actions according to the current policy and interacting with the environment over time, we can then obtain the state-action sequence trajectory as $\{s_0, a_0, s_1, a_1, \dots, s_T\}$.

Reward: At time slot t , when action $\mathbf{a}_t$ is chosen at state $\mathbf{s}_t$, the immediate reward received is the sum data rate of the networks r_t . Therefore, the discounted accumulated reward of a trajectory can be expressed as:

$$R_t = \sum_{i=0}^T \beta^i r_i, \quad (28)$$

where $\beta \in (0, 1)$ is the discount factor, which determines the effect of the current immediate reward on the future sum rewards. To ensure that the trained solutions satisfy AP capacity constraints (16) and (17), we assign a larger penalty when the number of devices connected to AP exceeds the limitation, thus forcing the agent to avoid infeasible solutions. The detailed S-PPO method is shown in Algorithm 1.

6. Performance evaluation

In this section, we evaluate the effectiveness of the proposed S-PPO-based solution algorithm through comprehensive simulations.

6.1. Simulation setup

We consider an indoor room area of $10 \times 10 \times 3.5 \text{ m}^3$, as illustrated in Fig. 1. To study the impact of inter-cell interference from LiFi APs on the

Algorithm 1: S-PPO for device association problem in hybrid LiFi/WiFi indoor networks.

Input: Initialize the parameters θ_0 of policy π (i.e., actor network), the parameters ρ_0 of the value function V (i.e., critic network).

Output: Optimal policy π^*

```

1 while episode < max episode do
2   while timestep < update interval do
3     for each device do
4       Select action  $a_k(t)$  according to policy  $\pi_{\theta_t}$ ;
5       Environment calculates reward  $R_t$  based on combined
        action  $\mathbf{a}_t$ ;
6       Observe new combined state  $\mathbf{s}_t$  based on local
        observations and previous experience;
7       Store experience tuple into memory;
8   Update policy  $\pi$  with the clipped surrogate objective (18);
9   Update value function  $V$  with the loss function (22);
10  Clear memory;
11  timestep  $\leftarrow 0$ ;
```

sum throughput of the hybrid network, two representative AP configurations are evaluated: interference-free scenario and interference-prone scenario, where the numbers of APs and devices range from 5 to 7 and 2 to 18, respectively. Example topologies are depicted in Fig. 4. Fig. 4(a) shows an interference-free scenario configuration, which employs a layout of $L = 4$ LiFi APs located at the four corners of the room, and $W = 1$ WiFi AP at the center. The circled areas with different colors indicate the coverage areas of different LiFi APs, while the WiFi AP covers the entire room. In contrast, the interference-prone scenario as presented in Fig. 4(b), has the same number of APs as the first scenario. However, in this scenario, the LiFi APs are positioned closer to each other to enhance coverage within the room. Consequently, it leads to overlapping coverage areas, increasing the likelihood of inter-cell LiFi interference. Without loss of generality, our proposed S-PPO method can be applied to other scenarios with different numbers of APs and devices. The parameters of LiFi and WiFi networks are summarized in Tables 1 and 2, respectively.

To validate the proposed method, two different network settings with different network capacity constraints are considered: (i) networking setting 1: $N_l = 2$, $N_w = 5$; (ii) networking setting 2: $N_l = 3$, $N_w = 7$. The device location and height are generated randomly. For each networking setting, the static device scenario and the mobile device scenario are studied, respectively. In the mobile device scenario, as discussed before, the RWP model is used, where the device moving speed is randomly selected between 0.5 m/s and 2 m/s. The device location information is obtained every 0.1 s, ensuring that the devices' locations are tracked in a nearly real-time manner yet do not change significantly. The RL parameters used in the simulation are summarized in Table 3.

Table 1
LiFi channel parameters.

Parameter	Symbol	Value
Height between ceiling and device	h	1.5 m - 2 m
The physical area of a PD	A	1 cm ²
The gain of the optical filter	$T_s(\psi_{i,k})$	1.0
Receiver FoV semi-angle	$\Psi_{1/2}$	90 °
Refractive Index	n	1.5
Half-intensity radiation angle	$\Phi_{1/2}$	60 °
Detector responsivity	κ	0.53 A/W
Transmit optical power per LiFi AP	P_{lf}	3 Watt
Noise power spectral density	$\mathcal{N}_{lf}$	10 ⁻²¹ A ² /Hz
Bandwidth per LiFi AP	B_{lf}	40 MHz

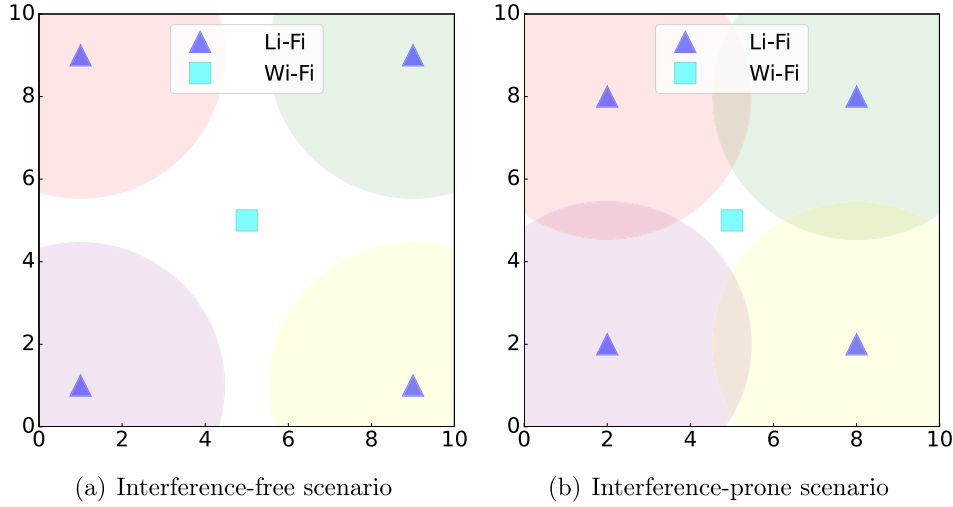


Fig. 4. Hybrid LiFi/WiFi networks topology.

Table 2
WiFi channel parameters.

Parameter	Symbol	Value
Breakpoint distance	d_{BP}	5 m
Central carrier frequency	f_c	2.4 GHz
Transmit power	P_{wf}	0.1 Watt
Bandwidth per WiFi channel	B_{wf}	10 MHz
Noise power spectral density	$\mathcal{N}_{wf}$	4.002×10^{-17} W/Hz

Table 3
S-PPO hyperparameters.

Parameter	Value
Learning rate	0.0001
Clipping coefficient (ϵ)	0.1
Discount factor (β)	0.99
Number of epochs per iteration	8
Update interval	450

6.2. Compared methods

Exhaustive search: Exhaustive search is a brute-force approach to solving optimization problems by evaluating all possible solutions. For the proposed problem, ES is used to enumerate all possible AP-device association cases to find the globally optimal AP-device associations, which will provide the upper bound of our work but at the cost of significant computational complexity. The objective function of exhaustive search is the same as (12).

Linear programming rounding: LPR is a commonly used approximation technique to solve the formulated BIP problem [39]. Problem 1 is first relaxed into a linear program by replacing the binary user-AP association variable with continuous values in $[0, 1]$. The relaxed problem is solved using the CVXPY framework, where both the ECOS and Gurobi solvers are employed separately to obtain the fractional solutions and compare their solution times. Finally, these fractional solutions are converted into feasible integer solutions through a rounding algorithm. The objective function of LPR is the same as (12).

Signal strength strategy: SSS is a widely adopted technique in wireless communication, focusing on selecting the AP that provides the highest signal-to-noise ratio for the devices [17,24]. When device k is served by AP i at time slot t , the received SNR $\sigma_{i,k}$ is denoted as:

$$\sigma_{i,k}(t) = \begin{cases} \frac{(\kappa H_{i,k}^{lf}(t) P_{lf})^2}{\mathcal{N}_{lf}^2 B_{lf}} & \text{if } i \text{ is LiFi AP,} \\ \frac{|H_{i,k}^{wf}(t)|^2 P_{wf}}{\mathcal{N}_{wf} B_{wf}} & \text{if } i \text{ is WiFi AP.} \end{cases} \quad (29)$$

Then, the objective function of SSS method for device k can be written as:

$$\begin{aligned} \max_{\mathbf{u}_k(t)} \quad & \sigma_{i,k}(t) \\ \text{s.t.} \quad & i \in \mathcal{L} \cup \mathcal{W}, \\ & k \in \mathcal{K}. \end{aligned} \quad (30)$$

Trust region policy optimization: TRPO is chosen for comparison due to its application in addressing the device-AP association problem within hybrid LiFi/WiFi networks [20]. It operates within the same structural framework as S-PPO, utilizing actor and critic networks. However, TRPO differentiates itself by implementing policy updates through trust region constraints, while PPO ensures stability using a clipped function. Compared to TRPO, PPO eliminates the need for complex second-order optimization and enables mini-batch stochastic gradient updates, resulting in improved computational efficiency and ease of implementation. The objective function of TRPO is the same as (12). The learning rate and discount factor for TRPO are the same as those for S-PPO.

Trust region policy optimization-domain knowledge: We further include the TRPO-DK in [27], which incorporates domain knowledge (DK) into the learning process. Specifically, instead of considering all candidate APs, it exploits the observation that the two highest-SNR APs are usually sufficient to serve the user demands. As a result, the action space is reduced, thereby decreasing the complexity.

6.3. Complexity analysis

As discussed in Section 5, the proposed S-PPO framework consists of actor and critic networks, with each being a 2-hidden-layer deep neural network, where each layer has N_h neurons. The computational complexity of the actor network to process one device is $\mathcal{O}(2(L+W)N_h + N_h^2)$. Then for the K device, it is $\mathcal{O}(K(2(L+W)N_h + N_h^2))$. While the critic network has a complexity of $\mathcal{O}((L+W)N_h + N_h^2 + N_h)$. Consequently, the aggregate computational complexity of the S-PPO methodology is calculated as $\mathcal{O}((2K+1)(L+W)N_h + (K+1)N_h^2 + N_h)$. Due to identical network architectures, the TRPO method shares the same complexity level as S-PPO. In contrast, the TRPO-DK approach incorporates domain knowledge by restricting the candidate APs to the two highest-SNR APs, leading to a complexity of $\mathcal{O}((2K+1)N_h + (K+1)N_h^2 + N_h)$. In comparison, the computational complexities for ES and SSS techniques for device-AP association are $\mathcal{O}((L+W)^K)$ and $\mathcal{O}((L+W)K)$, respectively. For the LPR method, solving the relaxed linear programming problem takes $\mathcal{O}(K(L+W)^3)$ time complexity, and the rounding process takes $\mathcal{O}(K(L+W))$ time [39]. Therefore, the overall complexity is dominated

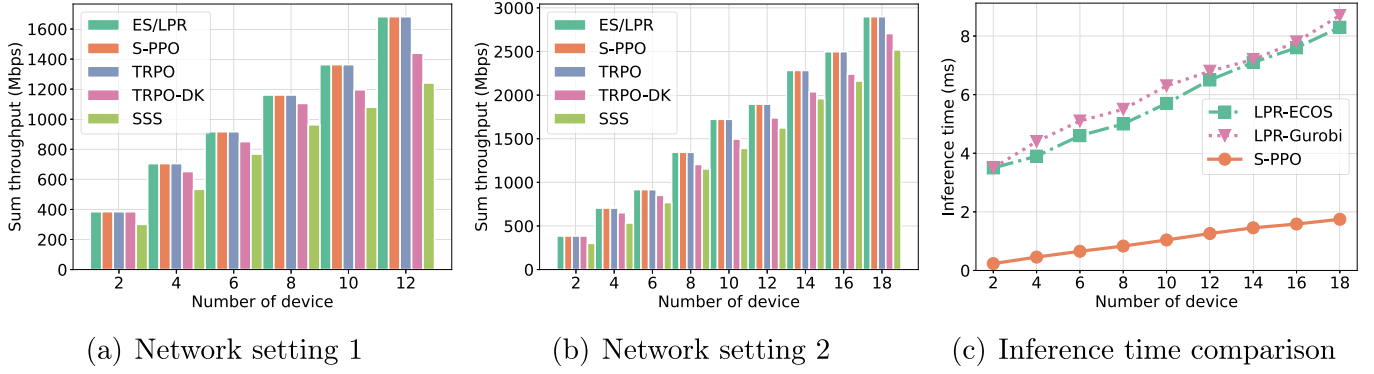


Fig. 5. Optimality comparison in terms of sum data rate in static interference-prone scenario.

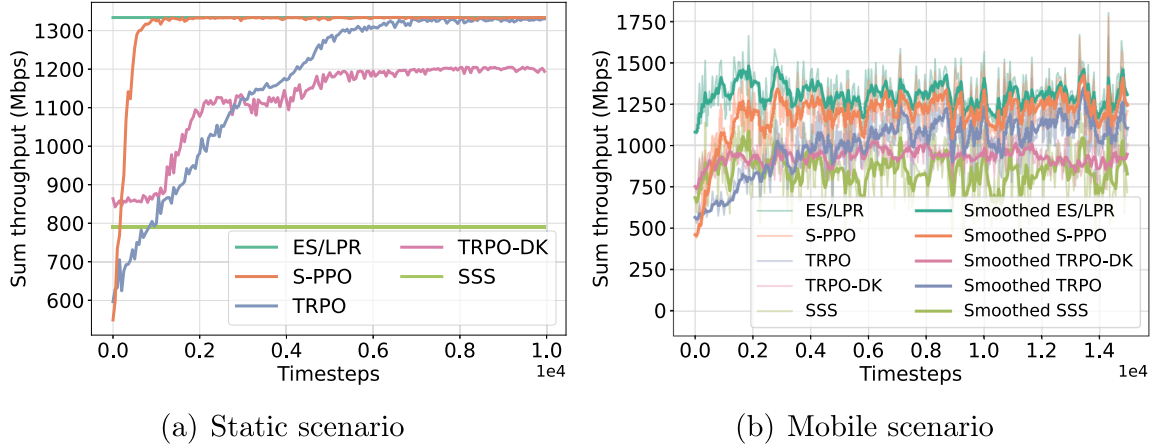


Fig. 6. Convergence comparison for 12 devices in interference-free scenario, network setting 2.

by the linear programming solving stage, resulting in a total time complexity of $\mathcal{O}((K(L+W))^3)$.

6.4. Simulation results

In this section, the performance of the proposed S-PPO method is evaluated by comparing it with the results obtained from ES, SSS, and TRPO in terms of optimality, convergence, robustness, computational complexity, scalability, and fairness under different network scenarios and settings. Additionally, we evaluate the fairness of the network.

Optimality: We first examine the optimality of the proposed S-PPO in static interference-prone networking scenarios with different capacity constraints by comparing the sum data rate with the baseline methods. As illustrated in Fig. 5(a) and (b), it is obvious that both the S-PPO method and TRPO can achieve the global optimal solution as the results achieved by the benchmark method ES. Moreover, S-PPO outperforms SSS and TRPO-DK by up to 35.5% in both network settings. It can also be observed that as the number of devices increases, the performance of the proposed PPO will not degrade, thus validating its reliability. It is worth noting that the LPR method can also achieve the global optimum. However, as shown in Fig. 5(c), the inference time of LPR with both the ECOS and Gurobi solvers is up to 15 times that of the well-trained S-PPO, whereas the inference time of S-PPO is only 1.746 ms even with 18 users.

Convergence: We then evaluate the convergence speed of the proposed S-PPO method under network setting 2 with 12 devices, considering both interference-free and interference-prone scenarios in static and mobile cases. In the static case, as Figs. 6(a) and 7(a) demonstrate, S-PPO converges 3 and 6 times faster than TRPO in interference-free and interference-prone scenarios, respectively. Moreover, both S-PPO

and TRPO methods can achieve the global optimum and outperform SSS and TRPO-DK by up to 16.7% with respect to sum throughput in the interference-prone scenario.

In the mobile scenario, to better illustrate the convergence trend, an exponential moving average is applied to the raw data. As depicted in Figs. 6(b) and 7(b), the proposed S-PPO method demonstrates significantly faster convergence, achieving up to a 300% improvement over TRPO. Moreover, S-PPO achieves near-optimal performance and outperforms TRPO, TRPO-DK, and SSS.

Robustness: To evaluate the robustness of the proposed S-PPO, we first conduct simulations to examine its sensitivity to changes in three important parameters, including learning rate (LR), clipping parameter, and user speed. Fig. 8(a) shows the impact of different learning rates on the performance of the proposed S-PPO. We can clearly see that the performance of S-PPO is not significantly affected in the particular range of LR from 0.00001 to 0.001. Though the performance is greatly degraded with a LR of 0.01, since it is not widely adopted, without loss of generality, we don't consider it a big drawback of the proposed S-PPO. The results of clipping parameters are shown in Fig. 8(b), where the clipping parameter ranges from 0.1 to 0.9 with an incremental interval of 0.2. The results remain consistent within the range, except for a light improvement observed at a clipping parameter of 0.1 compared with other values. Fig. 8(c) demonstrates the results under different constant user movement speeds. The results show that the proposed S-PPO method consistently achieves near-optimal performance and outperforms the three baseline methods. S-PPO outperforms the baselines by up to 32.16% even when the user speed reaches 2.0 m/s, while the gap to the optimum is only 8.48%. Based on the results discussed above, we can conclude that the proposed S-PPO method shows its robustness against changes in learning rate, clipping parameter, and user speed.

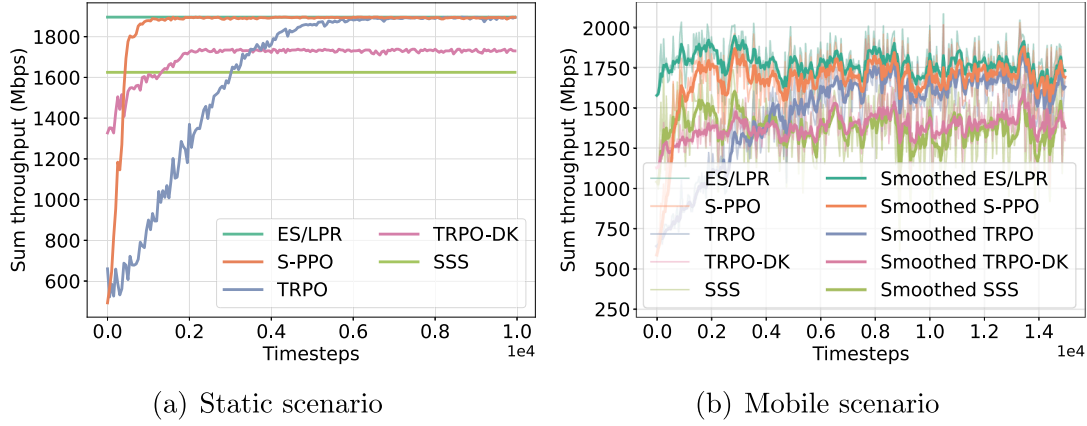


Fig. 7. Convergence comparison for 12 devices in interference-prone scenario, network setting 2.

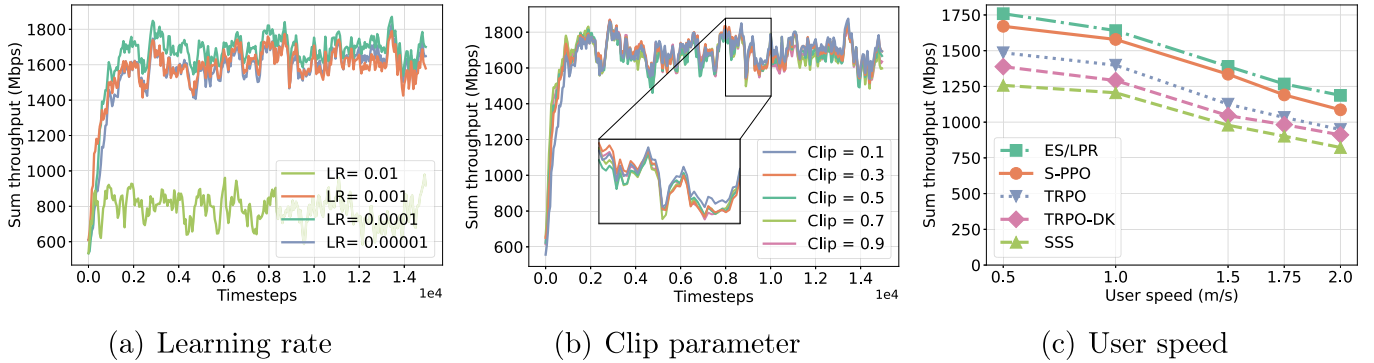


Fig. 8. Robustness comparison for 12 devices in interference-prone scenario, network setting 2.

Considering the vulnerability of the LiFi APs to blockage caused by obstacles, we next examine the robustness of the proposed S-PPO method in three blockage scenarios. (i) *Static blockage scenario*, as shown in Fig. 9(a), where a wall is located from (8, 0) to (8, 5) in the interference-free scenario, leading to a permanent LoS blockage in the shadowed area. Fig. 9(b) shows the results under static blockage with network setting 1, where the number of devices ranges from 2 to 12. It can be observed that as the number of devices increases, S-PPO achieves near-optimal results and consistently outperforms TRPO, TRPO-DK, and SSS. (ii) *Dynamic blockage scenario*, where time-varying obstacles stochastically block the LoS path between LiFi APs and user devices. As shown in Fig. 10(a), in the interference-prone scenario with network setting 1 and 20 % dynamic blockage rate, S-PPO achieves higher sum throughput than baseline methods as the number of devices increases. As expected, we also observe a slight reduction in sum throughput compared to the results in the same scenario but without blockage (Fig. 13(a)). Fig. 10(b) further illustrates the sum throughput performance for 8 devices in the network setting 2 with respect to different blockage rates. Though the sum throughput decreases as the blockage rate increases, the proposed S-PPO method consistently achieves performance near the global optimum obtained by ES. (iii) *Hybrid blockage*, where static and dynamic blockage coexist. Fig. 11(a) presents the robustness results in terms of hybrid blockage in the interference-free scenario with network setting 1. It can be clearly observed that as the number of devices increases, the sum data rate increases, with S-PPO outperforming the baselines by up to 35.4 %. Moreover, Fig. 11(b) illustrates the impact of different blockage rates in network setting 2 with 8 devices, showing that the proposed S-PPO consistently outperforms baseline methods. These results confirm the robustness of S-PPO in different blockage scenarios.

Computational complexity: We further evaluate the computational complexity pertaining to the training time of the proposed S-PPO

Table 4

Training time comparison in interference-prone scenario, network setting 2.

Number of device	S-PPO (s)	TRPO (s)
2	17.9	19.4
4	39.6	43.6
6	65.9	70.3
8	94.0	102.3
10	128.8	139.1
12	172.2	184.6
14	211.0	230.6
16	249.9	276.9
18	303.3	322.8

method by comparing it with that of TRPO, as they have the same network architectures. Specifically, we use a 2023 MacBook Pro with an Apple M2 Max chip, 32 GB of RAM, and a 1 TB SSD, operating under macOS Sonoma version 14.3. Python version 3.11.4 and PyTorch version 2.0.1 are used to implement the algorithms. The runtime comparison for the interference-prone scenario with network setting 2 is shown in Table 4. We can see that S-PPO consumes less time than TRPO across varying device numbers. It is important to note that even at a high device number of 18, the runtime of S-PPO is still within a feasible and acceptable range. This observation further indicates that the training time of S-PPO is relatively short, thus enabling its wide applicability in different network application scenarios. Moreover, as shown in Fig. 5(c), the inference time of S-PPO is only at millisecond-level, which demonstrates the practicality of S-PPO for real-time applications.

Scalability: To evaluate the scalability of the proposed S-PPO method, we then conduct extensive simulations by changing the number of devices from 2 to 18 and the AP number from 5 to 7 in different net-

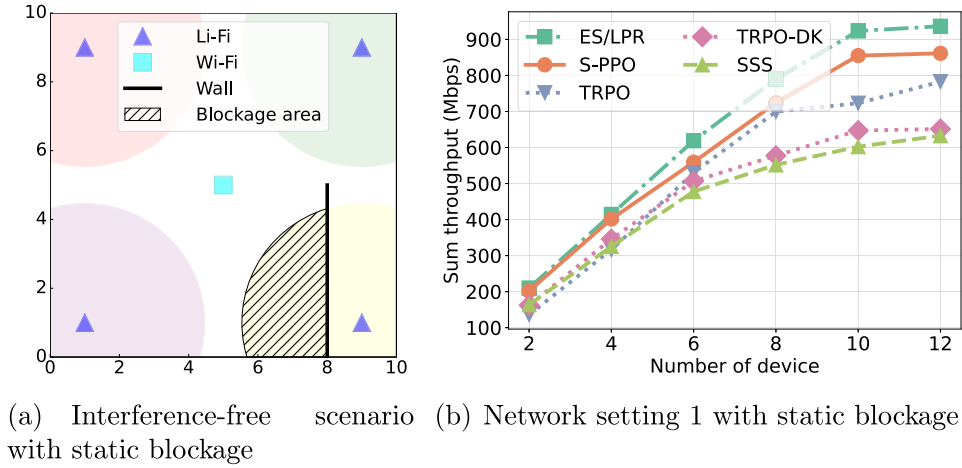


Fig. 9. Robustness comparison in terms of static blockage in interference-free scenario.

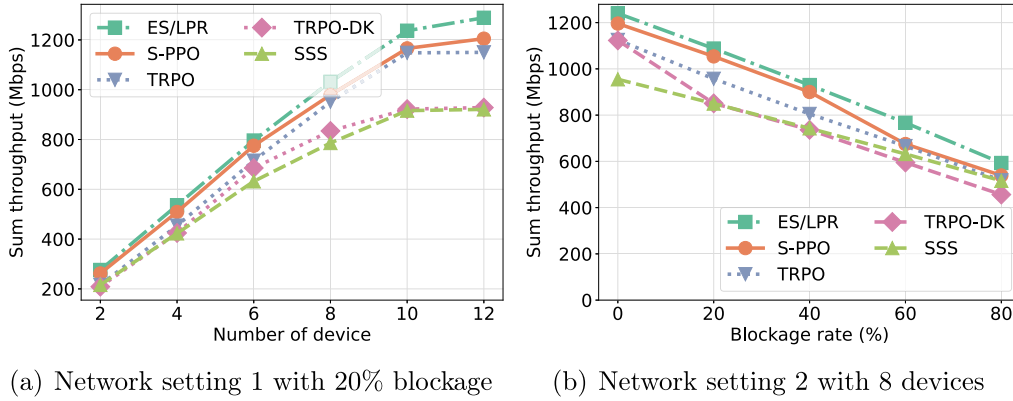


Fig. 10. Robustness comparison in terms of dynamic blockage in interference-prone scenario.

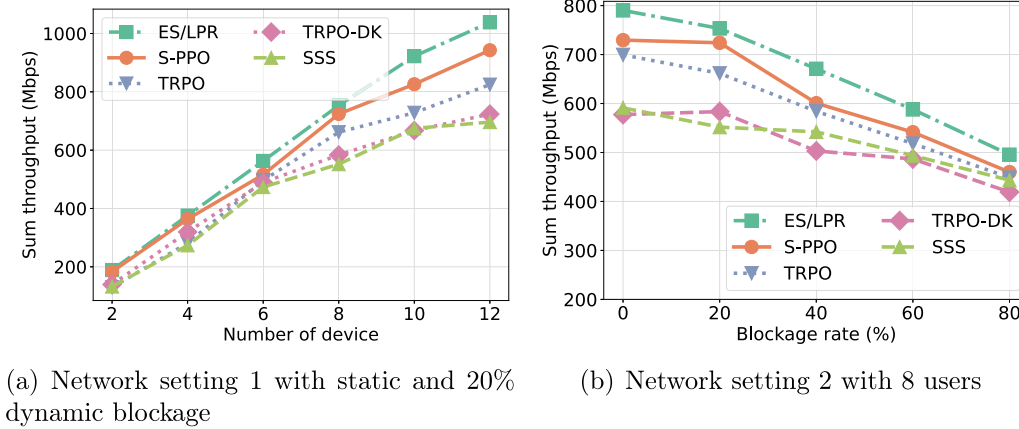


Fig. 11. Robustness comparison in terms of hybrid blockage in interference-free scenario.

work settings. Fig. 12 shows the scalability comparison in two network settings under the interference-free scenario. We can obviously observe that as the number of devices increases, the sum data rate increases. Moreover, compared with network setting 1 (Fig. 12(a)), network setting 2 (Fig. 12(b)) results in a higher sum throughput, particularly as the number of devices increases. This phenomenon is due to the increased capacity limitation in the network setting 2, where more devices can be accommodated. Moreover, S-PPO constantly outperforms TRPO, TRPO-DK, and SSS in both network settings, thus validating its scalability.

Fig. 13 illustrates the scalability results in the interference-prone scenario with respect to different numbers of devices. We can see that the proposed S-PPO method achieves nearly optimal results and outperforms TRPO, TRPO-DK, and SSS. Furthermore, compared to the results of the interference-free scenario in Fig. 10, the achievable sum throughput in the interference-prone scenario is higher in both network setting 1 and network setting 2. This is because the coverage area of the LiFi APs is larger in the interference-prone scenario, even with potentially increased LiFi inter-cell interference, a higher throughput can still be provided by wider coverage of LiFi APs.

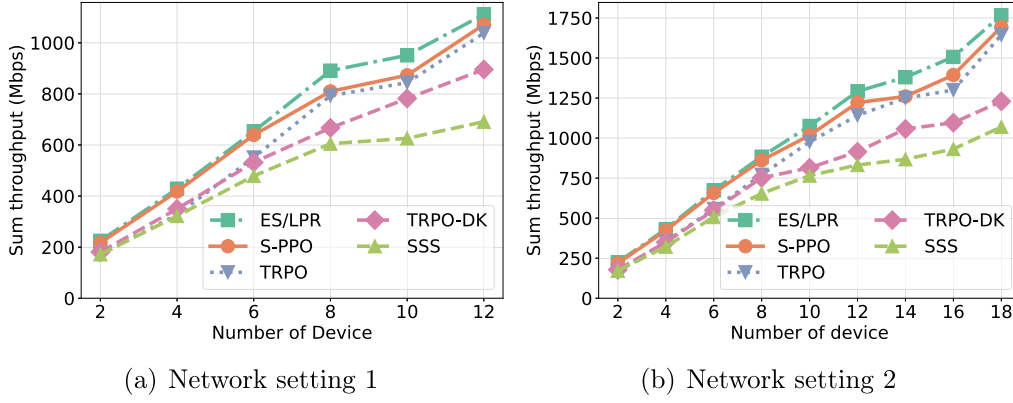


Fig. 12. Scalability comparison in interference-free scenario.

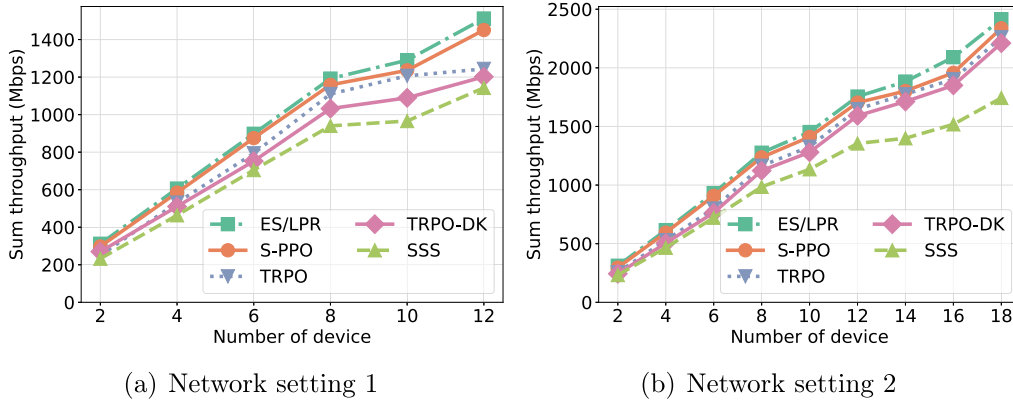


Fig. 13. Scalability comparison in interference-prone scenario.

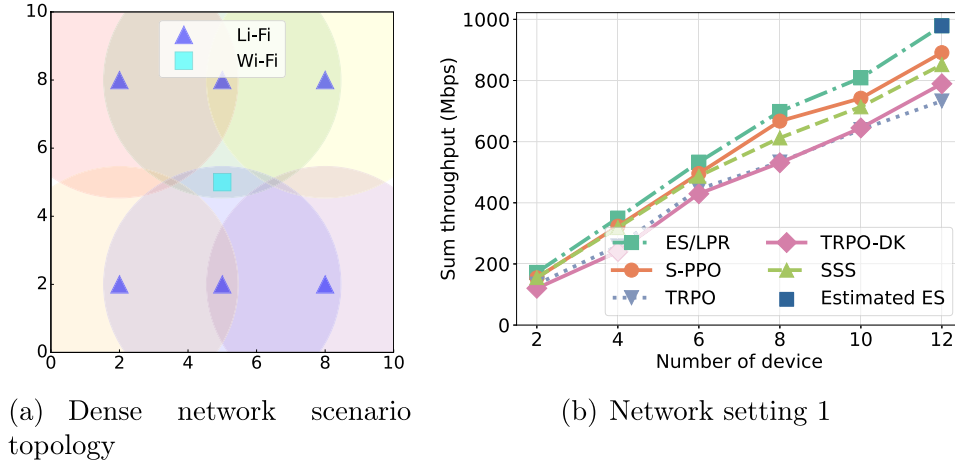


Fig. 14. Scalability comparison in dense network scenario.

We further explore the scalability in a denser network scenario, where the number of LiFi APs increases to 6, as illustrated in Fig. 14(a). The sum throughput comparison results in network setting 1 are shown in Fig. 14(b), where we can see that S-PPO can achieve 92.2% of the global optimal obtained by ES. The sum data rates denoted by pink markers are estimated values derived from the ES method because the computer cannot examine all the possible solutions within polynomial time. In this scenario, the sum throughput is not increased as expected. In contrast, the sum throughput decreases in all methods compared with that in the above-discussed 4-LiFi-AP interference-free

and interference-prone scenarios. This is because the denser deployment of LiFi APs significantly increases inter-cell interference, which adversely impacts the throughput provided by LiFi APs, thus leading to a lower sum throughput. However, the proposed S-PPO algorithm can still achieve better performance in terms of sum throughput than baseline methods.

In conclusion, the proposed S-PPO method can achieve nearly optimal performance with the increase of device number and AP number in different network settings and can outperform TRPO, TRPO-DK, and SSS.

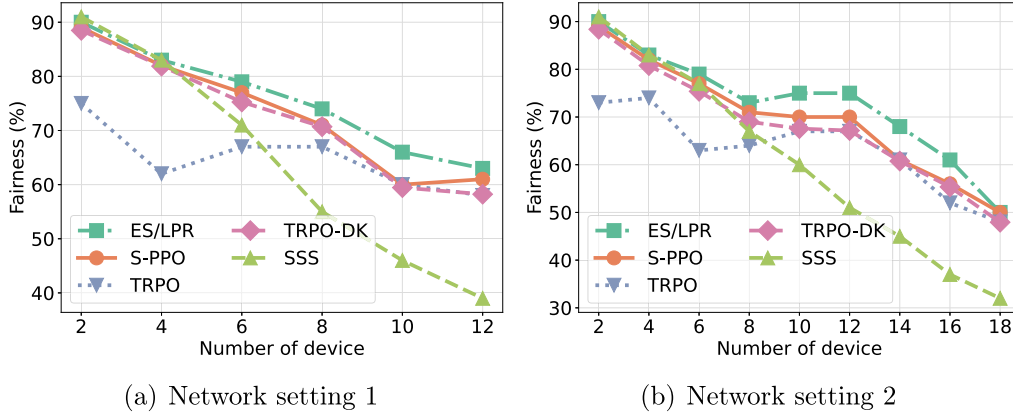


Fig. 15. Fairness comparison in interference-free scenario.

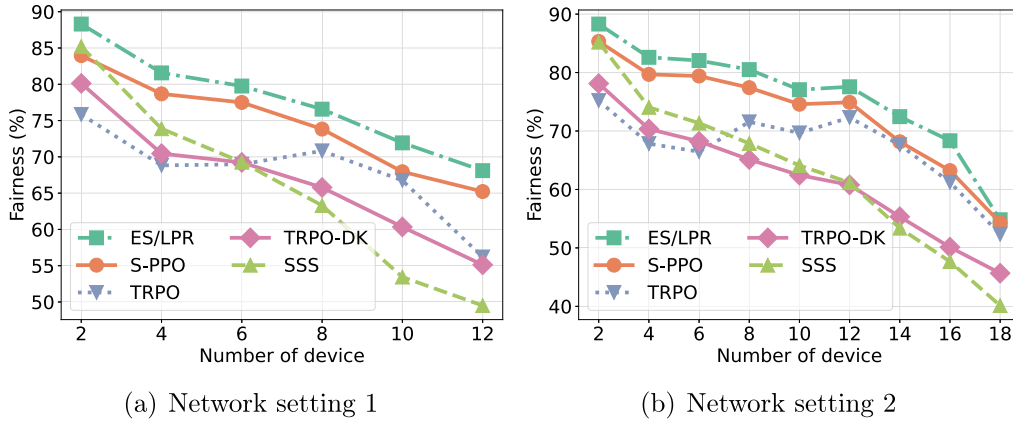


Fig. 16. Fairness comparison in interference-prone scenario.

Fairness: Besides the sum throughput of the hybrid LiFi/WiFi networks, we also evaluate the fairness of the network, where Jain's fairness index [40] is adopted, given as:

$$\xi = \frac{\left(\sum_{k=1}^K r_k / r_k^{req} \right)^2}{K \sum_{k=1}^K \left(r_k / r_k^{req} \right)^2}, \quad (31)$$

where r_k^{req} denotes the required data rate for device k . The required data rate is modeled as a random variable following a Gaussian distribution [24].

Figs. 15 and 16 show the results of fairness in the interference-free and interference-prone scenarios, respectively. As the number of devices increases, we can see that the fairness decreases. This is caused by the practical capacity limitation in hybrid LiFi/WiFi networks, where the growing demand due to the increasing number of devices cannot be completely accommodated. Therefore, the satisfaction ratio (r_k / r_k^{req}) decreases, thus resulting in a decrease in fairness. In addition, S-PPO outperforms TRPO, TRPO-DK, and SSS. Specifically, with only two devices, SSS exhibits greater fairness than ES in Fig. 16(a). This is because the objective of ES is to maximize the sum throughput, which doesn't guarantee optimal fairness. Therefore, SSS can obtain slightly better fairness than ES in this case. However, as the number of devices increases, the performance of SSS drops dramatically and is worse than the other three methods, because it cannot satisfy all the devices' required data rates due to limited capacity.

To further evaluate the optimality of the fairness that the proposed S-PPO can achieve, we modify the objective function only related to fairness as:

$$\text{Maximize } f = \xi(t) \quad (32)$$

$$\text{Subject to: (13), (14), (15), (16), (17).} \quad (33)$$

As depicted in Fig. 17(a), the proposed S-PPO method can achieve higher fairness but significantly lower sum throughput compared with that achieved by the proposed sum throughput optimization problem when the number of devices is less than 8. This may be due to the required device data rate may vary greatly when the number of devices is small. Therefore, maximizing fairness may sacrifice some devices' data rate, thus resulting in a lower sum throughput than what is achieved by focusing solely on maximizing throughput without considering fairness. As the number of devices increases and their data rate requirements become more uniform, a fairness-focused approach can achieve throughput similar to those obtained when the objective function is sum throughput, even though the results for fairness do not demonstrate significant improvement.

This phenomenon unveils the complex relationship between achieving fair resource allocation and maximizing overall network throughput. Relying solely on fairness as the objective function leads to inconsistent performance results, whereas our formulated objective function demonstrates both robust and consistent performance across various scenarios. In our future work, we plan to incorporate both fairness and sum throughput into our objective function to leverage the trade-off between them.

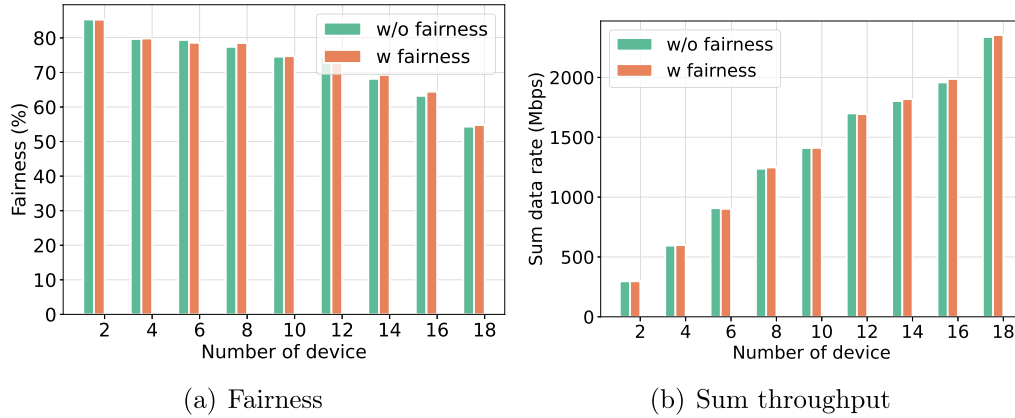


Fig. 17. Fairness-oriented optimization in interference-prone scenario, network setting 2.

7. Discussion

Deployment and real-time implementation. This work has demonstrated that the inference time of the proposed S-PPO achieves millisecond-level performance, enabling real-time device-AP association in dynamic environments. Additionally, this algorithm can be deployed on a central unit to gather network information and manage device-AP assignments. Therefore, this method is practical for real-world applications.

Fairness in the objective function. As mentioned in the previous section, while our work primarily focuses on optimizing the network sum throughput, other QoS metrics such as fairness are also essential for network performance. In future work, we plan to incorporate these QoS metrics into a multi-objective function. By balancing throughput, fairness, and other QoS factors, we aim to provide better device satisfaction and achieve load balancing across the hybrid networks.

Scalability to large-scale networks. In this paper, we have conducted extensive simulations to evaluate the scalability of the proposed S-PPO under different scenarios with up to 7 APs and 18 devices. Due to the limitations of devices in our lab, large-scale networks cannot be evaluated. However, distributed algorithms [41] and graph neural networks [42,43] are promising approaches for addressing large-scale scenarios, and we will leave it for future work.

Adaptability to varying AP and device numbers. In this work, we mainly focus on the scenarios with fixed AP and device numbers. However, in practical applications, the number of APs and devices may vary over time, such as when devices leave the network or new devices join. Therefore, an adaptive method is needed for the varying environment. In future work, graph neural networks could be explored as a potential approach to model and address the challenges posed by varying AP and device numbers.

8. Conclusions

We study and formulate the device-AP association problem in hybrid indoor LiFi/WiFi networks mathematically, where the network control objective is to maximize the network-wide throughput by associating devices to appropriate APs with consideration of device indoor mobility patterns and network capacity limitations. The resulting network control problem is a binary integer programming problem. We then propose the S-PPO method with the action space decomposition approach. Finally, we evaluate the proposed S-PPO method with extensive simulation results, comparing it with ES, LPR, TRPO, TRPO-DK, and SSS methods. Results show that the proposed solution algorithm can achieve 100% of the global optimum in the static device scenario, and converge 267% faster than TRPO in the mobile device scenario. We also validate the optimality, convergence, robustness, computational complex-

ity, scalability, and fairness of S-PPO in various scenarios. In the future, we plan to investigate the distributed algorithm to improve the scalability to large-scale networks and the adaptability to varying AP and device numbers. We will explore multi-objective optimization problem to leverage the tradeoff between different QoS metrics to further improve performance in hybrid LiFi/WiFi networks.

CRediT authorship contribution statement

Peijun Hou: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation; **Nan Cen:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Statista, Wide-Area and Short-Range IoT Device Installed Base Worldwide 2014–2029, 2022, <https://www.statista.com/statistics/1016276/wide-area-and-short-range-iot-device-installed-base-worldwide/>.
- [2] T. Komine, M. Nakagawa, Fundamental analysis for visible-light communication system using LED lights, *IEEE Trans. Consum. Electron.* 50 (1) (2004) 100–107.
- [3] N. Cen, J. Jagannath, S. Moretti, Z. Guan, T. Melodia, LANET: visible-light ad hoc networks, *Ad Hoc Netw.* 84 (2019) 107–123.
- [4] P. Hou, N. Cen, Proximal policy optimization for user association in hybrid LiFi/WiFi indoor networks, in: *Proc. of IEEE GLOBECOM*, Kuala Lumpur, Malaysia, December 2023.
- [5] Y. Zhang, N. Cen, Programmable software-defined testbed for visible light UAV networks: architecture design and implementation, in: *2023 IEEE 20th Consumer Communications & Networking Conference (CCNC)*, Las Vegas, USA, January 2023, pp. 843–848.
- [6] Y. Long, N. Cen, Q-learning For sum-throughput optimization in wireless visible-light UAV networks, in: *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, New York, USA, May 2023, pp. 1–6.
- [7] P. Pesek, S. Zvánovec, P. Chvojka, Z. Ghassemloo, P.A. Haigh, Demonstration of a hybrid FSO/VLC link for the last mile and last meter networks, *IEEE Photonics J.* 11 (1) (2019) 1–7.
- [8] A. Gomez, K. Shi, C. Quintana, M. Sato, G. Faulkner, B.C. Thomsen, D. O'Brien, Beyond 100-Gb/s indoor wide field-of-view optical wireless communications, *IEEE Photonics Technol. Lett.* 27 (4) (2015) 367–370.

- [9] X. Wu, M.D. Soltani, L. Zhou, M. Safari, H. Haas, Hybrid LiFi and WiFi networks: a survey, *IEEE Commun. Surv. Tutor.* 23 (2) (2021) 1398–1420.
- [10] M.B. Rahaim, A.M. Vegni, T.D.C. Little, A hybrid radio frequency and broadcast visible light communication system, in: *Proc. of IEEE GLOBECOM Workshops (GC Wkshps)*, Houston, USA, October 2011.
- [11] J. Kong, Z.-Y. Wu, M. Ismail, E. Serpedin, K.A. Qaraqe, Q-learning based two-timescale power allocation for multi-homing hybrid RF/VLC networks, *IEEE Wireless Commun. Lett.* 9 (4) (2020) 443–447.
- [12] H. Yang, A. Alphones, W.-D. Zhong, C. Chen, X. Xie, Learning-based energy-efficient resource management by heterogeneous RF/VLC for ultra-reliable low-latency industrial IoT networks, *IEEE Trans. Ind. Inf.* 16 (8) (2020) 5565–5576.
- [13] J. Chen, Z. Wang, Coordination game theory-based adaptive topology control for hybrid VLC/RF VANET, *IEEE Trans. Commun.* 69 (8) (2021) 5312–5324.
- [14] Y. Wang, X. Wu, H. Haas, Fuzzy logic based dynamic handover scheme for indoor Li-Fi and RF hybrid network, in: *2016 IEEE International Conference on Communications (ICC)*, Kuala Lumpur, Malaysia, May 2016, pp. 1–6.
- [15] J.P. Pavon, S. Choi, Link adaptation strategy for IEEE 802.11 WLAN via received signal strength measurement, in: *Proc. of IEEE International Conference on Communications*, 2, Anchorage, USA, May 2003.
- [16] D.A. Basnayaka, H. Haas, Design and analysis of a hybrid radio frequency and visible light communication system, *IEEE Trans. Commun.* 65 (10) (2017) 4334–4347.
- [17] X. Wu, H. Haas, Mobility-aware load balancing for hybrid LiFi and WiFi networks, *J. Opt. Commun. Netw.* 11 (12) (2019) 588–597.
- [18] J. Wang, C. Jiang, H. Zhang, X. Zhang, V.C.M. Leung, L. Hanzo, Learning-aided network association for hybrid indoor LiFi-WiFi systems, *IEEE Trans. Veh. Technol.* 67 (4) (2018) 3561–3574.
- [19] Z. Du, C. Wang, Y. Sun, G. Wu, Context-aware indoor VLC/RF heterogeneous network selection: reinforcement learning with knowledge transfer, *IEEE Access* 6 (2018) 33275–33284.
- [20] R. Ahmad, M.D. Soltani, M. Safari, A. Srivastava, A. Das, Reinforcement learning based load balancing for hybrid LiFi WiFi networks, *IEEE Access* 8 (2020) 132273–132284.
- [21] X. Wu, H. Haas, Load balancing for hybrid LiFi and WiFi networks: to tackle user mobility and light-path blockage, *IEEE Trans. Commun.* 68 (3) (2020) 1675–1683.
- [22] I.W.G. da Silva, E. Eduardo Benitez Olivo, M. Katz, D. Pamela Moya Osorio, Analysis and simulation of precoding and user association for securing hybrid RF/VLC systems, *IEEE Sens. J.* 24 (20) (2024) 33467–33480.
- [23] Y. Wang, X. Wu, H. Haas, Load balancing game with shadowing effect for indoor hybrid LiFi/RF networks, *IEEE Trans. Wireless Commun.* 16 (4) (2017) 2366–2378.
- [24] X. Wu, M. Safari, H. Haas, Access point selection for hybrid Li-Fi and Wi-Fi networks, *IEEE Trans. Commun.* 65 (12) (2017) 5375–5385.
- [25] H. Ji, X. Wu, Q. Wang, S.J. Redmond, I. Tavakkolnia, Adaptive target-condition neural network: DNN-aided load balancing for hybrid LiFi and WiFi networks, *IEEE Trans. Wireless Commun.* 23 (7) (2024) 7307–7318.
- [26] E. Guérin, T. Begin, A. Busson, I. Guérin Lassous, Towards a throughput and energy efficient association strategy for Wi-Fi/LiFi heterogeneous networks, in: *Proceedings of the 18th ACM Symposium on Performance Evaluation of Wireless Ad Hoc, Sensor, & Ubiquitous Networks*, Alicante, Spain, November 2021, pp. 119–126.
- [27] R. Ahmad, M.D. Soltani, M. Safari, A. Srivastava, Reinforcement learning-based near-optimal load balancing for heterogeneous LiFi WiFi network, *IEEE Syst. J.* 16 (2) (2022) 3084–3095.
- [28] M. Sana, A. De Domenico, W. Yu, Y. Lostonlen, E. Calvanese Strinati, Multi-agent reinforcement learning for adaptive user association in dynamic mmwave networks, *IEEE Trans. Wireless Commun.* 19 (10) (2020) 6520–6534.
- [29] J.M. Kahn, J.R. Barry, Wireless infrared communications, *Proc. IEEE* 85 (2) (1997) 265–298.
- [30] J.-B. Wang, Q.-S. Hu, J. Wang, M. Chen, J.-Y. Wang, Tight bounds on channel capacity for dimmable visible light communications, *J. Lightwave Technol.* 31 (23) (2013) 3771–3779.
- [31] E. Perahia, R. Stacey, *Next Generation Wireless LANs: 802.11 n and 802.11 ac*, Cambridge University Press, 2013.
- [32] D.B. Johnson, D.A. Maltz, *Dynamic source routing in ad hoc wireless networks*, in: *Mobile Computing*, Springer, 1996, pp. 153–181.
- [33] P. Hou, N. Cen, DC-PPO for joint user association and power allocation in dynamic indoor hybrid VLC/RF networks, in: *2024 IEEE 25th International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM)*, 2024, pp. 186–195.
- [34] A. Singh, H.B. Salameh, M. Ayyash, H. Elagla, Characterization of dynamic blockages for VLC-Enabled indoor industrial networks, *IEEE Syst. J.* 18 (4) (2024) 2173–2176.
- [35] S. Boyd, *Convex optimization*, Cambridge UP (2004).
- [36] V. Konda, J. Tsitsiklis, Actor-Critic algorithms, *Adv. Neural Inf. Process. Syst.* 12 (2000) 1008–1024.
- [37] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, (2017). [arXiv:1707.06347](https://arxiv.org/abs/1707.06347)
- [38] X. He, X. Zhuge, F. Dang, W. Xu, Z. Yang, DeepScheduler: enabling flow-aware scheduling in time-sensitive networking, in: *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications*, New York, USA, May 2023, pp. 1–10.
- [39] S. Paramita, A. Bhattacharya, R. Ahmad, V.A. Bohara, A. Srivastava, Flow-based rate maximization for link aggregation enabled hybrid LiFi-WiFi network, *IEEE Trans. Veh. Technol.* 74 (2) (2025) 3269–3282.
- [40] R.K. Jain, D.-M.W. Chiu, W.R. Hawe, et al., A quantitative measure of fairness and discrimination, *Eastern Research Laboratory, Digital Equipment Corporation*, Hudson, MA 21 (1984) 1.
- [41] S. Bayhan, E. Coronado, R. Riggio, A. Zubow, User-AP association management in software-defined WLANs, *IEEE Trans. Netw. Serv. Manage.* 17 (3) (2020) 1838–1852.
- [42] W. Deng, Y. Liu, M. Li, M. Lei, GNN-aided user association and beam selection for mmwave integrated heterogeneous networks, *IEEE Wireless Commun. Lett.* (2023).
- [43] H. Zhang, X. Ma, X. Liu, L. Li, K. Sun, GNN-based power allocation and user association in digital twin network for the terahertz band, *IEEE J. Sel. Areas Commun.* 41 (10) (2023) 3111–3121.